AIRSTREET · PROJECT PLAN

AirStreet project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/airstreet		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-30	Amish Chadha	Decisions 1 to 5 done on 2026-10-30; node cost USD 437.00
0.1	2026-10-24	Amish Chadha	First project plan with stage gates and the reference list (kit 1.9.1 upgrade)

AirStreet project plan

AirStreet is at stage P3 (design outputs, TRL 3): the constructable design, drawings, priced BOM, build plan, decisions register and first-pass design FMEA exist, and every concept option is modelled to D3. The next gate is P4 (verification, TRL 4), on hold while the portfolio is capped at TRL 3; Amish decided the five open decisions of AST-DEC-001 on 2026-10-30 (AST-DDR-006), so none is open now; the pole owner's load confirmation and the approach to the first partner are his actions. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Design and CAD	Design lead	D3 model of every option, STEP files, drawings AST-DWG-001 and 101 to 109	Done at P3 (kit 1.9.1)
Calculations and STEM check	Design lead	AST-CAL-001 v0.8, AST-STM-001	Done; twelve plausible rows of option A to close at TRL 4 or in FieldNode
Requirements and traceability	Design lead	AST-URS-001, AST-PRS-001, AST-RTM-001	Done; targets to review with the partner
Build plan	Design lead	AST-BLD-001	Done (plan, not yet built)
Partner and collocation site	Project lead	Agreement with a city or agency that runs a street-level NO2 reference station	Not started; first candidate named (AST-DEC-001)
FieldNode interface	Project lead, with FieldNode	FieldNode's agreement to the adapter plate mount and the always-on port B	Request approved 2026-10-02; agreement outstanding
Renders and storefront	Project lead	Options picture, product renders and cards from `cad/src/product_model.py`	To render on Amish's Mac

Resources needed

Table 2. Resources (TRL 4 work, on hold).

Resource	Deliverable	Effort	Estimated cost (USD)
Parts for two prototype nodes and one spare pod	Built nodes for collocation	Purchase	About 1,000 (two nodes at 437.00 plus a pod at about 130; estimate)
3D printer time and a maker	Printed saddles, pods, shields	About 3 days	Material only (in the BOM)
CalRig time	T, RH and PM checks per pod	About 2 days	Lab time
Reference station collocation	14 days before deployment, then every 6 months	Partner time	To agree with the partner
Installer with a work platform	Two installations and one exchange	Half a day each	To quote locally

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Done
P1 Definition	URS, PRS draft, RSK with hazard analysis, MKT	Done
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR, STEM check closed	Done except the STEM check (open: seven plausible rows of option A) and the concept selection (decided by Amish on 2026-10-30)
P3 Design outputs	Model, STEP, drawings, priced BOM, build plan, DEC, design FMEA, traceability	Done (kit 1.9.1, 2026-10-24)
P4 Verification	Lab and collocation tests (TST), verification report	On hold (TRL cap 3)
P5 Validation	Field trial with the partner	On hold
P6 Commercialization	Only with a named deployment partner	On hold

Next steps and decision requested

1. Done on 2026-10-30: Amish decided the five decisions of AST-DEC-001 (AST-DDR-006): option A, the diffusion tube cross-check at the anchor node and one node in five, flagged hot hours now, both EU limits for PR-004, and the text of PR-017 and PR-018.
2. Render the options picture and the product renders on Amish's Mac (`python .kit/options.py` ; `.kit/export_views.py` , `.kit/photoreal.py` , `.kit/cards.py`).
3. Approach the first partner candidate; when agreed, fix the radio band and FieldNode antenna in the BOM.
4. When the TRL cap is lifted: build two nodes to AST-BLD-001 and collocate them (P4).

References

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